

# AURA: Automated Analysis and Reporting of Therapeutically Applied Table-top Role-Playing Games

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Figure 1: A character sheet and a set of polyhedral dice used to play Table Top Role-Playing Games (TTRPGs).  
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## Abstract

There is a global shortage of therapeutic, educational, and social support for neurodivergent children and youth, with substantial downstream consequences for individual well-being and societal costs in adulthood. While early and sustained support is known to be both humane and economically effective, existing service models struggle to scale. Therapeutically Assisted Role-Playing Games (TARPGs) have recently emerged as a promising, group-based intervention that combines guided play, narrative structure, and social skill rehearsal to support neurodivergent development. Case studies consistently report positive outcomes, including increased engagement, social connection, emotional regulation, and self-efficacy. However, current research remains dominated by small-scale and non-experimental case studies, and practical barriers persist for wider adoption, especially for carers without formal therapeutic training. Therapists are often not available, and carers are the next, and more numerous, line of support for neurodivergent children.

This paper introduces AURA (AI-Assisted Understanding of Role-play), a novel system designed to support therapists and carers running TARPG sessions. AURA addresses a key practical challenge in TARPG facilitation: documenting complex, multi-player narrative sessions without disrupting live interaction. By analysing audio and/or video recordings of sessions, AURA generates structured post-session summaries, and key event timeline, enabling facilitators to focus on participants rather than note-taking. AURA aims to lower barriers to entry, extend TARPG capacity beyond scarce clinical settings, and support more scalable, evidence-informed interventions for neurodivergent children and youth.

## CCS Concepts

• **Applied computing** → **Health informatics**; • **Computing methodologies** → **Information extraction**; *Speech recognition*; • **Human-centered computing** → *Interactive systems and tools*.

## Keywords

tabletop role-playing games, neurodivergence, therapeutic games, speech-to-text, summarization

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## 1 Introduction

There is a global shortage of therapeutic, educational, and social capacity for supporting neurodivergent children and youth, limiting access to activities and interventions that foster social, emotional, and cognitive development [17, 21, 60, 74, 75, 77, 81, 85, 86]. This, in turn, can lead to a variety of challenges later in life, across psychological, social, and financial aspects of adult life [3, 33, 53, 61, 83, 87].

*Therapeutically Assisted (or Applied) Role-Playing Games* (TARPGs) provide a promising approach towards increasing therapeutic support of neurodivergent clients, especially for children and youths, but also adults [6, 10, 41, 45, 55, 66]. In these interventions, participants assume characters and co-create narratives, rehearsing social, emotional, and problem-solving skills in a safe, imaginative setting [6, 22, 39, 41, 46, 65]. TARPGs have the potential to provide a keystone component in a larger strategic framework towards addressing the gap of support resources for children and adolescents with neurodevelopmental differences (again, here understood broadly to cover neurodivergence) [6, 10, 41, 45, 55, 66].

There are a number of challenges to consider when introducing TARPGs as a tool for therapists, as well as when extending the reach of TARPGs beyond them to non-therapist *carers* (e.g., parents, social workers) more generally. These include bespoke training, support, and specialized game systems such as *Critical Core* [25]. Furthermore, there is a need for game facilitator (or Game Master, GM, the person managing the game) support systems to address the practical challenge of keeping track of events, taking notes about player actions and developments, and observing player behaviour. Such note-taking and observations can be highly intrusive to the flow of a TARPG session and require substantial training, which can be a factor in limiting broader uptake. Additionally, when running TARPGs, there is also a need for recording and observing the actions and development of the participating players, necessary for post-session analyses. However, there are currently no specialized tools available for running TARPGs towards supporting such analyses, which can make this process cumbersome [22, 39, 46].

## 2 Contribution

Here we introduce the AURA system as a first step towards solving the challenge of supporting TARPG game facilitators (GMs) by keeping track of game sessions and providing automated capture, transcription, condensation and analysis of TARPG sessions. Figure 2 demonstrates an output of AURA, compared to the relevant segments of the raw transcript and a human-written summary. Developing systems that can record, analyse, and report on TARPG sessions to support game facilitators (whether therapist or carer, e.g. parent, teacher, social worker) is complicated by the highly complex nature of TTRPGs (whether played face-to-face at a tabletop or remotely online). Due to the length of transcripts and the inherent complexities in TTRPG narrative structures and participant communication, this is not a trivial task. Moreover, as detailed by, e.g., [26, 31, 42, 78], keeping track of events in these games can be highly challenging, exhibiting complexity beyond, e.g., analysing the narratives of television and cinema [18].

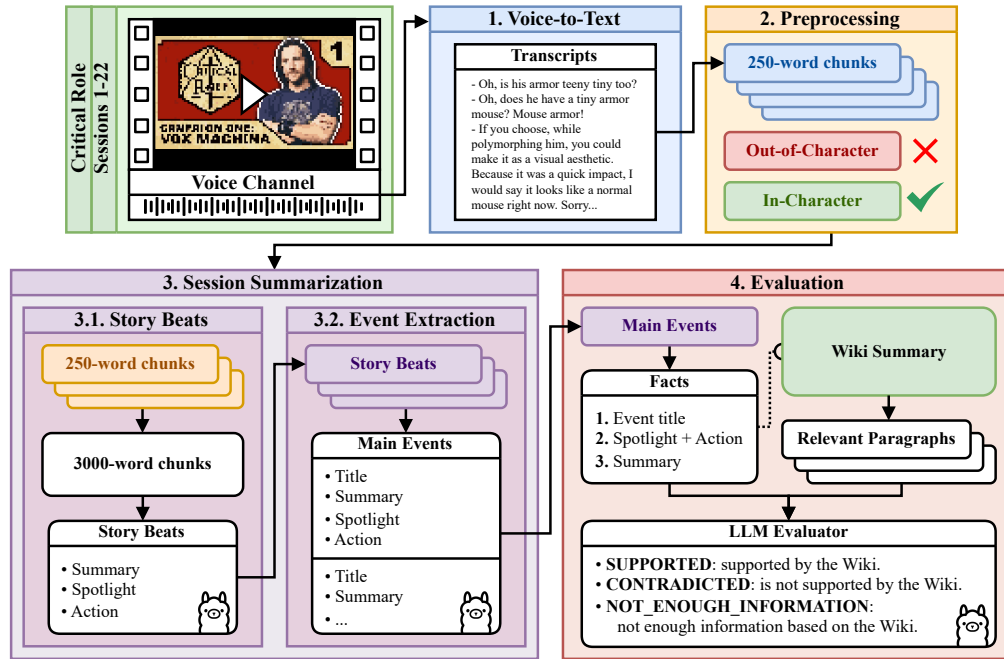
AURA produces structured post-session reports, summarizing the narrative, providing an overview of key events. The system is designed around the current need from therapists and carers

| Critical Role: Campaign One, Session One |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Transcript                               | Arcanist Allura Vysoren of the Tal'Dorei council came to them requesting their aid saying that a long time friend of hers, Lady Kima of Vord, who is a very well known and very well respected halfling paladin of Bahamut, the Platinum Dragon, had been gone on a pilgrimage for a while, essentially a vision quest that is part of her own development as a paladin. As part of this, she let the information go to Allura that a dark vision had come to her saying that some sort of evil root is beginning to breed beneath Kraghammer and the mountains within Kraghammer, [...] However, she managed to acquire the necessary documentation and offered you a very substantial reward should you find the whereabouts of Lady Kima of Vord and hopefully bring her back safely. |  |
|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
| Wiki Summary                             | <b>Arrival at Kraghammer</b><br><b>First night</b><br>Vox Machina have just entered the dwarven city of Kraghammer for the first time on their quest to find Lady Kima of Vord. They know she was headed for the mithral mine owned by Nostoc Greyspine, but they decide to find somewhere to sleep first.<br>...                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
| AURA Summary                             | <b>A Dark Vision and a Plea for Help</b><br>Lady Kima, a respected halfling paladin of Bahamut, warns the party of an evil root breeding beneath Kraghammer and asks for their aid in finding her whereabouts.<br><b>Spotlight: Lady Kima</b><br>She requests the aid of the party in finding her whereabouts and bringing her back safely.                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |

**Figure 2: Example of the AURA output. Top: excerpt from the beginning of the first session transcript. Bottom left: the first paragraph of the Wiki summary of the session, taken from [criticalrole.fandom.com](https://criticalrole.fandom.com). Bottom right: the first event in the AURA summary, roughly corresponding to the transcript and human-written summary.**

to focus on the game and the players (participants) while also making notes and documenting progress for clinical purposes, or for informing future game sessions. AURA is intended to lower the barrier to entry and is intended to allow therapists and carers (and others) to focus on active sessions as opposed to note-taking and observation.

Figure 3 shows the high-level overview of the proposed pipeline and its evaluation method. Throughout the paper we use *Critical Role, Campaign One* [23] as an example, however, the pipeline designed to be game agnostic. The prototype of AURA operates on the voice recording of play sessions. We use a voice-to-text algorithm to generate written transcripts of the session (see Section 5.2). These transcripts are cleaned of discussion not pertinent to the game (see Section 5.3). The cleaned transcripts are processed in two steps using Large Language Models (LLMs). First, they are turned into short story-beats. Then we generate a list of meaningful events from these beats (see Section 5.4). The final summary contains these events, along with the most important—or “spotlight”—character in the segment and their most important actions. As Fig. 2 illustrates, the output of this pipeline is a list of concise events, important characters, and their role in the ongoing story. To check the accuracy of the summaries, we use human-written recaps as ground truth. We find the most relevant paragraphs in the recaps using the cosine similarity of the text embeddings of both the recaps and our generated events. We give these paragraphs to a large LLM to evaluate the truthfulness of the generated events (see Section 6.3.2).



**Figure 3: Overview of the AURA pipeline for TTRPG Session Summarization and Summary Evaluation.** 1. The voice channel from sessions from *Critical Role: Campaign One* are transcribed using a voice-to-text algorithm. 2. The transcripts are preprocessed in chunks; using a zero-shot text classifier, out-of-character chatter is removed. 3. The sessions are summarized in two passes: 3.1. transcript chunks are aggregated, and an LLM summarizer is used to generate Story Beats for each chunk; 3.2 the list of Story Beats are then passed to another LLM to generate the list of Main Events in the session. 4. The results are evaluated by a larger LLM that compares the facts extracted from the Main Events list with the most relevant paragraphs of human-written summaries from the fan Wiki.

### 3 Background: Neurodivergence and TARPGs

Neurodivergence typically refers to a variety of conditions and diagnoses (e.g., Autism Spectrum Disorder (ASD), Attention Deficit Hyperactivity Disorder (ADHD) as well as dyslexia and dyspraxia) [17, 21, 60, 74, 75, 77, 81, 85, 86]. The term neurodivergence is part of the neurodiversity framework. In this framework, neurodiversity refers to the diversity in the human brain leading to differences in the way people perceive, understand, and navigate their world, including ASD. Neurodivergence is used as a broad umbrella term for individuals in whom this variation diverges from that of the majority of the population, often termed neurotypical. This divergence is considered part of natural variation, and not deficient [71].

Research is clear in documenting, in exhausting detail, the current lack of support for neurodivergent children and youth [82], not the least around the “services cliff” as people move from childhood to adulthood [43, 72]. There is now a substantial body of research showing that children and adolescents with neurodevelopmental differences (ASD, ADHD, learning disabilities/neurodivergence more broadly) face elevated risks of poorer outcomes in adulthood when their supports are weak [3, 33, 53, 61, 83, 87]. Apart from the humanitarian problem of having a minority of the population facing such challenges, the financial impact of needing to support adults who did not receive the needed support while young is substantial. This can, in return, have a compounding effect on the immediate

family or carers [15, 16]. There is considerable evidence for the value of early support and intervention economics [15, 36, 47]; but also a gap in the support network and structures around neurodivergent children and youth. Ultimately, lack of support can lead to severe downstream effects, whether young or adult, as evidenced, e.g., by the elevated rates of neurodivergence among homeless people in the UK [34, 87]. As rates of neurodivergent diagnosis increasing, there is an acute need for solutions at all stages of the development chain from young child to adult. This even more acute given the backdrop of strain on public mental health support [15, 47].

**Therapeutically Assisted Role-Playing Games:** TARPGs are a form of multiplayer collaborative Role-Playing Games (RPGs). RPGs are incredibly varied in form and format, but can essentially be viewed as a type of role-playing bound by rules. Originating in table-top wargaming in the 1970s as well as oral storytelling traditions, improvisational theater, and elsewhere, RPGs have also inspired several videogame genres and variants, referred to as Computer Role-Playing Games (CRPGs) [26, 42]. The game form is incredibly varied, from single-player to multi-player, and has an extensive history (e.g., [24, 26, 31, 42]). Today, playing RPGs across any format is a globally distributed hobby [26]. Perhaps the most well-known RPG format is the Table-Top RPG (TTRPG), from which the idea of using RPGs in therapeutic contexts springs [22, 39, 46]. However, in practice, a variety of physical and digital RPG formats have

been adapted for therapeutic, educational, of developmental contexts [2, 5–8, 10–13, 19, 29, 41, 45, 50, 54, 55, 59, 65–67]. Similar to TTRPGs, TARPgs are collaborative games where participants assume characters and jointly create stories within structured rules. The therapist (or counselor) takes on the role of *story facilitator* or *game master*, *GM* [80], which generally means providing the information about the fictional world and its environment, inhabitants, as well as managing rules, arbitration, etc. [26, 42, 80].

**TARPgs as Interventions:** In recent years, various formats of TARPgs have emerged as a modality for interventions in a variety of neurodivergence contexts as well as more broadly in the context of education and mental health interventions (e.g., Post-Traumatic Stress Disorder (PTSD) treatment) [6, 10–12, 41, 45, 54, 55, 66]. Researchers, therapists, and *carers* (defined here broadly as any non-therapist support person, e.g., teachers, social workers, professionals in daycare or after-school programmes, parents) involved with neurodivergent children are increasingly interested in harnessing the playful, engaging format to support youth with, for example, addressing social, emotional, or developmental challenges.

TARPgs are promising across objective and subjective improvement measures [6, 11, 12, 41, 45, 54, 65, 66]. They combine guided play and narrative therapy, allowing young people to practice social interaction, empathy, and emotional regulation through imaginative play. This safe, structured format can engage even reluctant participants, fostering peer connection and transferable life skills [6, 41, 45, 55, 65, 66]. For therapists, RPGs provide a group-based format: they engage clients who may be reluctant to participate in traditional talk therapy, foster peer connection, and create opportunities for skill rehearsal that can generalize to real life. In economic terms, TARPgs enable one therapist to work with a group of children, rather than one at a time, increasing capacity in the mental health support system [6].

**The Role of Carers:** While TARPgs run without a professional therapist may not be therapy—the game is a tool for therapy, not therapy in itself—there are strong indications from the current case studies state-of-the-art of knowledge about TARPgs, the proliferation of the [76], and practitioner guidelines [22, 39, 46], to suggest that some level of developmental support of neurodivergent children and youth can be achieved by non-therapist **carers**, such as parents, social workers or teachers, running such games. However, there is a lack of formal, clinical comparative research on across TARPgs used by therapists vs. carers, but current case studies report positive outcomes in both situations: TARPgs provide an accessible tool to support children that scales support beyond scarce clinical settings. Their clear rules, shared storytelling, and cooperative play make them possible to facilitate even without formal therapeutic training. This may not be as transformative as if clinical therapists were managing the game, but the potential increase in support capacity is an incentive for the research in TARPgs and the development of new TARPg systems and solutions. This is crucial as there are more carers than professional therapists specialized in neurodivergent children and youth. If TARPgs could be used to systematically empower carers to extend developmental support, thus complementing therapeutic services, a significant scaling in

the support structures around neurodivergent children and youth could be the result.

## 4 State-of-the-Art

The academic state-of-the-art of knowledge around TARPgs is relatively well described across a few recent reviews and overview articles (e.g., [6, 10, 51, 65, 66]), but it needs to be noted that it is possible that a significant part of the collected applied knowledge is not captured in these academic articles, but rather is found in the communities worldwide where (TT)RPG interests and the need to support neurodivergent children and youth (or other vulnerable groups) overlap. The recent books published by therapists using TARPgs in their clinical practice (e.g., [22, 39, 46, 67], see also [13, 50]), and the origin story of *Critical Core* highlight that such practice can occur in a variety of contexts which may not be documented in academic literature. These sources also highlight the role of serendipity in leading academics, professionals, or carers to the idea of using TTRPGs for therapeutic or other supporting purposes. While keeping these gaps in the accessible written record in mind, this section will provide a condensed overview of that record, referring readers to the following sources for additional information: [6, 10, 22, 39, 46, 51, 66, 67].

Support systems, such as AURA, for the therapist or carer running a TARPg are not explicitly mentioned in existing work, but do exist implicitly in some contexts. For example, the use of maps and miniatures to create a visual representation of an encounter in a physical TTRPG game, or the use of virtual desktops for a remote TTRPG session. Virtual desktop tools for TTRPGs come with the ability to save maps and other tools, and possibly even with the ability to record a game session. This can also be done using standard screen recording software. Following, there are also tools available for automatically transcribing such a recording, although the ability of commercial tools to transcribe TTRPG sessions is currently not documented. Advice on note-taking and observation is also described in the publications from practicing therapists in the domain [22, 39, 46, 67].

### 4.1 TARPg Research

Reviews of TARPg research (e.g., [6, 11, 51]) and empirical research articles on TARPgs converge on promising but preliminary results, with the studies examined for this paper reporting subjective or objective improvements on a range of factors, including greater social connectedness, resilience, and self-confidence, as well as the capacity of TARPgs to reduce social anxiety. Previous research encompasses TTRPGs, as well as computer-based RPGs (e.g., remote play), and includes children, adolescents, and adults across various contexts. While this is a good indication of the flexibility of TARPgs across contexts, it does mean less empirical evidence for any given context or purpose. Furthermore, only a minority of previous studies employed experimental or even quasi-experimental designs, and form case studies. This is in line with what would be expected of an emerging clinical but cross-disciplinary domain. TARPgs are still characterized as being in an emerging state [6, 11, 51], but augmented by case-based experience from clinical therapists (e.g., [22, 39, 46]).

Arenas et al. [6] analysed 50 case studies on TARPGs, noting that most studies were published by 2011 (78%), indicating that the practice of therapeutically applied RPGs is not entirely new (see also e.g., [13]). The authors note that tabletop variants account for 44% of the studies, and digital 50%, both used as therapeutic or preventive interventions, typically incorporating cognitive-behavioral therapeutic approaches. Around 70% of the reviewed studies targeted adolescents, emphasizing the potential of these games as a complementary tool in psychotherapy. However, the authors note considerable variance in the definitions used for RPGs, the outcomes measured, and the interventions, which prevents a systematic review. Furthermore, Arenas et al. [6] highlighted that only 16% of the involved studies were quasi-experimental or experimental trials (e.g., [55]), which makes it difficult to draw clinical empirical conclusions based on the current state-of-the-art. Yuliawati et al. [51] explored the potential benefits of TTRPGs as an intervention across 51 publications in research. 35 were concluded to be review and qualitative design studies, with 12 being of quantitative design studies and 4 mixed-methods design studies. The authors conclude that existing literature highlights the benefits of applying TTRPGs as an intervention method. Benefits include the promotion of cognitive and psychosocial skills, as well as the prevention of negative effects and stress, and also as an aid to intervene in psychological problems such as social anxiety, depressive symptoms, and ASD. Notably, the review highlights the demonstrated strength of TARPGs across cognitive performance, specifically in enhancing creativity, problem-solving skills, and perspective switching [46].

On a further note, Yuliawati et al. [51], echoing a similar conclusion by Arenas et al. [6], note the ability of TARPGs to operate as an inclusive space for players to express their emotions without fear of consequences, which can help facilitate players in expressing themselves more openly to therapists (counselors). This not only benefits players, but also therapists as the game serves as a platform for them to observe patterns in behavior and trains of thought among the participating players. This can help therapists identify areas where players need further support [66].

Psychological interventions are generally grouped into *promotive*, *preventive* or *therapeutic* tools, TARPGs applied in the context of neurodivergent children appear to operate across these categories, but are typically described as a therapeutic tool [6, 11, 12, 41, 45, 51, 54, 55, 66]. This terminology is thus adopted here.

## 4.2 Reported Benefits of TARPGs

The reported benefits of RPG participation have been wide-ranging – for instance, increased motivation and self-efficacy, creativity, empathy, well-being, social connection, and sense of belonging have all been associated with RPG programs in various studies. These positive findings align with case reports from clinicians. For example, recent publications by therapeutic practitioners highlight the benefits of using RPGs such as TTRPGs in therapeutic contexts [22, 39, 46]. A recurrent motivation across the reports from therapeutic practitioners is that they are themselves players of TTRPGs (or other RPG formats), and realized that there was more to these games than simple entertainment. This is also the originating motivation behind the *Critical Core* TARPG game system, which was developed by neurodivergence researchers and therapists working

with the *Game to Grow* foundation [76]. *Critical Core* is the first toolset specifically focused on assisting therapists as well as carers getting started with using TARPGs to support autistic children (as well as neuronormative children). It is scaffolded around the Developmental Individual-Difference, Relationship-Based (DIR) (also referred to as the “*Floortime*”) model [38, 62] which identifies the relevant areas of functioning and helps with the construction of developmental profiles of children with nonprogressive developmental and learning disorders, including autism spectrum disorders (ASDs), and provides a developmental framework for the assessment and intervention process. Formally, the game system was adapted from the *Dungeons & Dragons* TTRPG rules system (5th edition) [76]. *Game to Grow* has also launched a programme training therapeutic GMs. Such training is also provided by companies such as *GeekTherapy* [35].

A further benefit of TARPGs focuses on the reluctance of some clients to engage in traditional therapeutic practices [51]. The authors highlight the resistance towards treatment in adolescents and adults, and in opening up and engaging fully in counseling sessions. There can be many reasons for this, including financial barriers or social isolation. In response to these challenges, Yuliawati et al. [51] and others (e.g., [6, 22, 39, 41, 46, 65]), highlight the imperative to explore alternative approaches that can engage clients in informal, more comfortable manners, highlighting the introduction of gamification in therapeutic interventions (across digital and analog formats, e.g., [52, 69]).

## 4.3 State-of-the-Art: Synthesis

Recent review studies and various case studies highlight the immature but highly promising nature of the evidence base. While applied in a range of contexts, TARPGs are gaining recognition for their ability to support participants facing a range of mental health challenges. TTRPGs create an immersive space where participants can develop coping skills, explore their identity, and practice emotional regulation [2, 11, 12, 41, 54, 65]. In group settings, particular benefits have been reported for participants with ASD, ADHD and special needs. Because TARPGs are structured activities [6, 20], Rubin-Budick [68] utilized TARPG scenarios to enable a better understanding of the perspective of children on the autism spectrum. Similar efforts are reported by e.g. [6, 65]. However, beyond the need for more experimental evidence, there are a number of gaps in the current state-of-the-art:

- (1) **TARPGs beyond professional therapists:** The current state-of-the-art is mainly focused on situations where professional therapists are involved in the application of TARPGs. There is less documentation available about the benefits when carers are facilitating these games.
- (2) **Challenges with cultural alignment:** Yuliawati et al. [51] note that a key limitation is the lack of research on the cultural alignment of TTRPG systems and approaches. Yuliawati et al. [51] suggest that TARPGs should be aligned with the cultural context of the involved players, as this can influence perception of TARPGs as well as the reactions of participants to rules, themes, and characters. There is a lack of culturally localized systems and scenarios.

- (3) **Barriers due to complexity:** Running a TARPg requires preparation. Added to this are practical barriers that can prevent uptake in a mental health support system that is already under-resourced [22, 39, 46]: For example, note-taking during dynamic play is intrusive, and no validated support tools exist to assist with the analysis or structuring of sessions, during run-time or post-session.
- (4) **TARPgs may not be suitable for everyone:** The benefits of TARPgs may be limited to participants who enjoy the gaming process and engage with the fictional setting of the game in question. Yuliawati et al. [51] also highlight that the underlying rules systems in TTRPGs can translate psychological elements into numbers and dice rolls [6, 65]. This can simplify or restrict the communication skills of a character and lead to discomfort among neurodiverse players [1].
- (5) **Fantastical settings can be a risk for some players:** Most previous case studies of TARPgs utilize a fantasy-style setting, eschewing the option to run a TTRPG even in a more realistic setting. This can be a risk for some players. For example, those who are currently experiencing psychosis or have other safety concerns may not be well-suited participants for TARPgs. Furthermore, immersion in a fantasy setting may distort players' perceptions of the real world [32].

In summary, the state-of-the-art positions TARPgs as a highly promising yet under-researched modality. A diversity of studies demonstrates potential to improve social and emotional outcomes for neurodivergent youth, while digital RPGs provide proof-of-concept for scalable therapeutic play.

#### 4.4 The Structure of Narratives and Play

The general nature of how TTRPGs operate, including many details, is covered in the existing literature (e.g., [26, 31, 42, 70, 78]). However, the details of the unfolding narrative is less well documented.

While there are a great variety of ways a TTRPG can be played, the traditional view is that of a group of players and a game master (story facilitator), jointly creating a shared narrative. Each player controls one character (*Player Controlled Character*, PC), the GM controls the shared, imagined world and the *Non-Player Controlled Characters* (NPCs) in that world. The game takes place in a description-response cycle where the GM provides information about something happening in the state of the world, and the players then decide how to react to it. As the GM and the players continually describe the actions of characters and NPCs, and the world state updates, a dynamically shaped story takes shape, which is partly observable and recordable, and partly unobservable, taking place only in the mind of each individual participant. The observable narrative and the perception of that narrative of each participant can be different to a greater or lesser extent, as each participant interprets events and encounters that have to be imagined differently [26, 31, 42, 57, 70, 78]. The game rules, or system, acts to maintain a general balance of power between the PCs and the world and its NPCs, and provides a structured framework for the players to have their characters act within the fictional environment.

At a surface level, an observed TTRPG session will, in many cases, be structurally similar to narratives told in other media formats, seeing PCs move between scenes or encounters. For example,

a game session may see the PCs meet at a tavern, travel to a castle, travel to a dungeon, explore the dungeon, and return to the castle. Within each geographical “scene”, a variety of encounters can take place, e.g., combat, social interaction, exploration, or similar [27, 31, 42, 57, 70, 78]. This view is, however, a simplification. For example, TTRPGs can demonstrate intricate temporal behaviours, where the characters controlled by the players are distributed in not just space but also time [79]. The participating players generally have a high degree of freedom to control their characters (PCs) within the shared, imagined world, allowing them to operate independently and affect the overall unfolding narrative. One of the key challenges of a GM is to ensure that the mental image all participants hold of what is going on in the game is sufficiently similar to enable collaborative storytelling. Additionally, the traditional understanding of scenes as creating structure in, e.g., literature or cinema, does not translate well to TTRPGs, where the transition between different narrative events can be fuzzy or even vary from player to player [26, 27, 31, 78, 79]. Jointly, compared to a strictly linear narrative with all characters following the same timeline, as would be observed in a film-based narrative [18], a TTRPG session is challenging to parse and describe, even for human observers. For example, there may not be clear transitions between scenes or encounters. This presents a concrete challenge for any automated system intended to capture narrative progression, events, or player/character actions in a TTRPG.

Additional challenges for automated transcription of TTRPGs (and therefore TARPgs) rest in the players communicating both as themselves and as their characters. The relationship between character and player is important therapeutically [22, 39, 46], but it is not necessarily stated clearly by a player whether they are speaking as themselves or as their character. Furthermore, the GM can play many roles and even vary their voice accordingly. Significant levels of overtalk—multiple people speaking at the same time, about potentially narrative events taking place across different times and places—further pose a challenge to automated systems attempting to transcribe or analyse TARPg sessions.

### 5 Methodology: Approach, Data, and Models

The core of the AURA system is essentially a session-summarizer. Here, we present the pipeline from capturing voice recordings, through voice-to-text transcription, to automatic summarization. Due to the length of the transcripts and the inherent complexities of TTRPG play, as detailed above, the task is not trivial. We approach the problem sequentially. The downstream task of our algorithms is to create a list of major events that happened during the session. We load and interface with our models through the Ollama<sup>1</sup> framework. We use the following system prompt: “*You are a JSON generator. You must respond ONLY with JSON that matches the provided schema. Do not include explanations.*”; and leverage Ollama’s capability to enforce JSON outputs.

#### 5.1 Play Sessions and Summaries

We source play sessions from the *Critical Role actual play* web show (produced by Geek & Sundry Inc.)<sup>2</sup>. To test our pipeline,

<sup>1</sup><https://ollama.com/>

<sup>2</sup><https://www.youtube.com/@GeekandSundry>

we focus on their first recorded campaign: *Campaign One: Vox Machina, Arc 1: Kraghammer and Vasselheim*. The game takes place in a fantasy setting, created by the game master, *Matthew Mercer*. Critical Role uses *Dungeons & Dragons 5th Edition* (*Wizards of the Coast*, 2014) as the rule set to govern the game. The sessions were recorded between March and September, 2015. The first arc consists of 22 episodes totaling 82 hours, 24 minutes, and 35 seconds of audio, with the average session running around 3 hours, 45 minutes long. Each session features the same game master and a recurring cast of seven players. In the later sessions, there are occasional appearances from one of four guest actors. While the sessions were produced for the viewers' entertainment, they are a continuation of a private game and very much improvisational. This first arc of the Critical Role web series arguably presents a realistic "optimal" TTRPG scenario where the players always focus on the story, the out-of-character chatter is kept at a minimum, and the game master is driving the game forward at a good pace. The main limitation of the recordings is the production environment itself, rather than the game content. Because the commercial nature of the web show, each session is split into two segments with an advertisement break and community announcements happening in between. However, because of the improvisational nature of the recordings, there is no set space for breaks, and intermissions happen at somewhat organic break points.

To verify our pipeline, we are using the session summaries available on the fan-made *Wiki* of the web show<sup>3</sup> [84]. While these articles contain meta information about the sessions, we focus only on the session synopsis. We extract paragraphs pertinent to the current session from these summaries (discarding sections on recaps and announcements). We treat these human-written *Wiki* entries as ground truth when we verify the factuality of our own summaries.

## 5.2 Voice-to-Text Transcription

We use whisperX [9] to transcribe the voice recordings. Using the audio stream of each episode as an input, whisperX generates a transcript and clusters spoken phrases into several speakers. We configure English as the default language for transcription to match the language of the play sessions. For each sentence, we store the identified speaker, the text, and the beginning and ending timestep.

WhisperX [9] and its underlying base model, Whisper [63], are known to struggle with transcribing overlapping speech. An inspection of our transcription outputs (see 6.1) confirms that such errors do occur, but they are largely confined to rare instances in which multiple players speak simultaneously about the current situation. We do not apply any post-processing to correct errors in these segments; however, this issue could be mitigated in future work by recording each player with a separate microphone.

## 5.3 Preprocessing

As a preprocessing step, we divide the transcripts into up to 250-word consecutive segments. We consider sentence boundaries when creating the segments, thus most segments are a few words shorter than 250. We use entailment prediction to discard segments with out-of-character chatter. We use a model based on *DeBERTaV3* [40] that has been fine-tuned on a large dataset of natural language

inference tasks [73]<sup>4</sup>. We set up the model with "entailment" and "contradiction" labels and the following premise: "*This text is out-of-character commentary by the player, not dialogue, not in-world narration, and not part of the roleplay story.*". We discard segments that entail this premise. The segment length is chosen to provide the model with sufficient context to reliably classify the segment type, while minimizing the risk that segments containing both in-character and out-of-character material remove substantial portions of the story.

Across all sessions, we discard between 0.83% – 9.64% ( $M = 4.74\%$ ) of the transcript segments. While this approach allowed us to remove sections with more obvious out-of-game content, such as commercial promotions, and community announcements, e.g.:

"So also as a heads up, because I saw some of you guys asking in the chat room, these videos will be available on the Geek & Sundry YouTube channel soon enough, as well as a more extensive backstory to the party [...]"

However, some of this content is hard to capture because of the thematic nature of the commercials. E.g. the following section is not picked up because "wizards" are categorized as "in-world narration":

"- We want all of you guys to go find these books, find either people in your town or online, and play.  
- And send pictures of your group to us. We'd love to see you guys play together.  
- Tag us, tag Wizards of the Coast, say what's up.  
- Yeah, tag Wizards of the Coast. Tell them these people are cool. They think they're cool. [...]"

## 5.4 Summarization

For the summarization tasks, we use the 3-billion parameter *Llama 3.2*<sup>5</sup> a light-weight text-only version of the Llama 3 foundation models [37], which was developed for on-device summarization along with other instruction-following tasks. We chose this model because our goal is to integrate the presented framework into a mobile application that will provide support for game masters. Leveraging small on-device models can improve the usability of applications and alleviate future security concerns.

For summarization, we first aggregate the preprocessed segments into larger chunks and derive summaries in multiple rounds. Based on a preliminary sensitivity analysis, we aggregate the cleaned chunks into up to 3000-word consecutive segments. These chunks only contain the text transcripts. We do not provide the speaker labels in the summarization pipeline; the character names are inferred from the transcript itself. In the first stage, we instruct the LLM to identify key story-beats in every chunk, with a focus on goals, conflicts, discoveries, twists, and consequences. Along with the main story beats, the LLM is also prompted to extract the most important character in the scene, and recap the actions made by this "spotlight" character. As an example, the first session is divided into ten chunks. From the last chunk, we extract two beats:

<sup>3</sup><https://criticalrole.fandom.com/wiki>

<sup>4</sup>[https://huggingface.co/knowledgator/comprehend\\_it-base](https://huggingface.co/knowledgator/comprehend_it-base)

<sup>5</sup><https://ai.meta.com/blog/llama-3-2-connect-2024-vision-edge-mobile-devices/>

**Summary:** The party encounters a strange naga beast and Keyleth is poisoned by its venom.

**Spotlight:** Keyleth, Scanlan

**Action:** Scanlan uses his bardic abilities to inspire himself and then casts Lightning Bolt at the naga beast, killing it.

**Summary:** The party discovers that the naga beast was a monster that had been emerging from a tunnel, and they learn about its weaknesses.

**Spotlight:** None

**Action:** One of the dwarves pats Scanlan on the shoulder and congratulates him on his victory.

The compiled story beats are then concatenated and sent to the LLM again. This time, we prompt the system to compile the major story events for the entire session based on the list of story beats. Similarly to the previous step, we keep track of the spotlight characters and their main actions. The final output is a list of events with a generated title, one- to two-sentence summary, spotlight character, and main actions. Figure 2 shows the output of the final summary.

## 6 Results

As discussed in Section 5.1, we evaluated our approach using publicly available videos of pen-and-paper sessions. Thanks to massive fan effort, episodes are well documented. The Wiki for the web show provides a mostly complete transcript of each episode as well as a high-level summary of characters, locations, and plot. These will be used as a ground truth for the remaining evaluation. Given that the compiled data from the wiki does not consist of information on the players' emotional state and personal highlights, we only demonstrate the outcomes of our pipeline for these stages.

### 6.1 Transcription

Using whisperx [9], we extracted a transcript for each video using the whisper-large-v3 model [63]. Table 1 shows the Word Error Rate (WER) [58] and Character Error Rate (CER) [44] for each episode to evaluate the quality of the resulting transcripts.

With an average WER of 25.4% and an average CER of 21.3%, the quality of the transcripts is enough to preserve the story and the discussions among participants. A closer inspection of the resulting transcripts reveals three kinds of common error sources:

- **Multiple players speak at the same time:** given that WER and CER are based on the Levenshtein distance (the number of required changes on a word-by-word basis of two texts) [48], the order in which parallel speech is transcribed can have a great impact on the evaluation scores. E.g., if two players A and B speak at the same time, the text could be reasonably transcribed in two orders, either first A, then B, or the other way around. In case the fan and the ML transcription have the same order, the error rates for this text section are 0. If different orders are chosen, the error rate is 1, while both are arguably representing the text correctly.
- **Hallucination of text passages:** given that whisper-large-v3 is a transformer model, it may hallucinate text passages during the transcription processes. Manual inspection of the

text revealed that sections of silence are rarely denoted by text that has not been said by the players.

- **Wrong ground-truth:** on rare occasions, the fan-written are not representing correctly what has been said by the players. E.g., during dialogues between two players, comments by other players are sometimes neglected. Whisper often picks up these short utterances and adds them to the transcript.

### 6.2 Speaker Annotation

As a second step, we use pyannote-audio [14] (which is part of the whisperX pipeline) to identify the speaker of each sentence of the whisperX-generated transcript. We use homogeneity, completeness, and v-measure [64] to compare the labels of the fan transcripts and the pyannote-audio speaker annotation. As the transcripts do not perfectly match, the cluster comparison of the two speaker annotations is made on a word-by-word basis. Further, only words that match between the two transcripts are considered to minimize the impact of errors in the previous transcription stage. Table 2 shows the quality of the speaker annotation per episode.

High scores in homogeneity ( $M = 0.71$ ) show that pyannote-audio can identify a matching speaker cluster for large parts of the transcript. However, lower values in completeness ( $M = 0.55$ ) indicate that a single cluster rarely covers all the sentences of a single person. The following typical error sources have been identified by us:

- **Professional voice-actors:** the players of Critical Role are professional voice-actors. Especially, the dungeon master Matthew Mercer tends to use different voices for each NPC. The transcript often identifies them as being spoken by him but not as a separate character.
- **in/out-of-character conversations:** while the other players typically stick to a single character they play, they still vary their voice between in- and out-of-character play. Similarly, this is not always indicated by the transcript.

These problems could be alleviated by using separate microphones per player or information from the video stream. As this is not the focus of our work, we leave this problem to be studied in future work.

### 6.3 Session Summarization

Once sessions are transcribed, the next step involves LLM-based summarization of the sessions. The overall process is described in Section 5.4. The goal of the summarization is to provide the game's facilitator with structured notes on the narrative content of the session. AURA outputs a list of events with a generated *title*, *summary*, identified *spotlight* character and their main *action(s)*. An example of the output can be seen on Fig. 2. Extracted story segments for each episode are available in the supplementary materials.

**6.3.1 Issues with Traditional Summary Metrics.** To verify the truthfulness of this output, we use an LLM evaluator. We have considered more traditional metric, such as the Recall-Oriented Understudy for Gisting Evaluation (ROUGE)-L [49] and BERT-Score [88], but these were found insufficient. ROGUE-L checks for the longest common subsequence. Because this score relies on lexical overlap, it is an unfair metric for evaluating greatly condensed summaries (such as

**Table 1: Evaluation of transcription quality based on Word Error Rate (WER) and Character Error Rate (CER) per episode.**

| Episode | WER   | CER   | Episode | WER   | CER   | Episode | WER   | CER   | Episode        | WER          | CER          |
|---------|-------|-------|---------|-------|-------|---------|-------|-------|----------------|--------------|--------------|
| 1       | 0.211 | 0.160 | 7       | 0.239 | 0.201 | 13      | 0.197 | 0.151 | 19             | 0.402        | 0.365        |
| 2       | 0.195 | 0.159 | 8       | 0.173 | 0.134 | 14      | 0.340 | 0.301 | 20             | 0.193        | 0.148        |
| 3       | 0.291 | 0.255 | 9       | 0.168 | 0.130 | 15      | 0.328 | 0.284 | 21             | 0.366        | 0.324        |
| 4       | 0.169 | 0.127 | 10      | 0.200 | 0.159 | 16      | 0.196 | 0.157 | 22             | 0.266        | 0.221        |
| 5       | 0.319 | 0.272 | 11      | 0.252 | 0.208 | 17      | 0.373 | 0.329 | 23             | 0.424        | 0.383        |
| 6       | 0.209 | 0.173 | 12      | 0.126 | 0.092 | 18      | 0.209 | 0.165 | <b>average</b> | <b>0.254</b> | <b>0.213</b> |

**Table 2: Evaluation of speaker clustering quality based on homogeneity (h), completeness (c), and v-measure (v) per episode.**

| Episode | h    | c    | v    | Episode | h    | c    | v    | Episode | h    | c    | v    | Episode        | h           | c           | v           |
|---------|------|------|------|---------|------|------|------|---------|------|------|------|----------------|-------------|-------------|-------------|
| 1       | 0.71 | 0.50 | 0.58 | 7       | 0.74 | 0.60 | 0.67 | 13      | 0.71 | 0.56 | 0.63 | 19             | 0.64        | 0.50        | 0.57        |
| 2       | 0.63 | 0.48 | 0.55 | 8       | 0.71 | 0.65 | 0.68 | 14      | 0.53 | 0.61 | 0.57 | 20             | 0.78        | 0.51        | 0.62        |
| 3       | 0.68 | 0.55 | 0.61 | 9       | 0.75 | 0.57 | 0.65 | 15      | 0.76 | 0.61 | 0.68 | 21             | 0.74        | 0.54        | 0.62        |
| 4       | 0.75 | 0.62 | 0.68 | 10      | 0.65 | 0.52 | 0.58 | 16      | 0.70 | 0.53 | 0.60 | 22             | 0.69        | 0.49        | 0.58        |
| 5       | 0.70 | 0.55 | 0.62 | 11      | 0.69 | 0.53 | 0.60 | 17      | 0.71 | 0.51 | 0.59 | 23             | 0.60        | 0.41        | 0.49        |
| 6       | 0.73 | 0.59 | 0.65 | 12      | 0.78 | 0.66 | 0.72 | 18      | 0.66 | 0.52 | 0.58 | <b>average</b> | <b>0.70</b> | <b>0.55</b> | <b>0.61</b> |

our case). Moreover, research has found that ROGUE scores do not align well with human evaluations [30]. BERT-Score is a measure of semantic similarity. Unfortunately, this measure works best with more diverse documents. Because all of our data comes from a relatively homogeneous source (the same players, playing the same continuous game), we found that BERT-Score gives uniformly high values to all of our output.

**6.3.2 LLM-based Evaluation Pipeline.** Due to the above mentioned difficulties with more traditional metrics, we turn to a large LLM to evaluate the AURA output. We use *Llama 4 Scout* [4], a state-of-the-art 109 billion parameter model trained for multimodal inference and reasoning. *Llama 4 Scout* uses a Mixture-of-Experts architecture, with 16 experts each with 17 billion active parameters at inference time. Our evaluator checks whether the list of AURA events is factual by comparing it to the Wiki-summaries introduced in Section 5.1. First, inspired by *FactScore* [56], we generate a list of atomic *facts* from our event titles, summaries, and spotlight character and action pairs. Then, we divide the ground truth summaries into paragraphs. Using a small embedding model<sup>6</sup> to measure the cosine similarity of each paragraph to the event in question, we fetch the top five closest paragraphs. Finally, we prompt the evaluator LLM to compare the facts to these paragraphs, considering action and events. The LLM provides a verdict of “supported”, “contradicted” or “not enough information” alongside a justification and evidence from the provided human-written paragraphs for the judgment, e.g.:

**Fact:** Spotlight character: Lady Kima. Main action: She requests the aid of the party in finding her whereabouts and bringing her back safely.

**Evidence:** Vox Machina have just entered the dwarven city of Kraghammer for the first time on their

quest to find Lady Kima of Vord.

**Justification:** The event states that Lady Kima is the spotlight character and the main action is that she requests aid in finding her whereabouts and bringing her back safely. The source evidence mentions that Vox Machina are looking for Lady Kima, as they ask Adra about a halfling and learn that Lady Kima was headed for the mithral mine owned by Nostoc Greyspine. This indicates that the party is indeed seeking her and trying to find her location, which aligns with the event.

**Verdict:** Supported.

**6.3.3 AURA Summary Results.** Table 3 shows the final results of the quality of AURA’s output. We break down our results based on the different types of atomic facts we generate from the events. On the event level, we consider events either supported or contradicted if at least 3 of the 4 facts generated from them have the same verdict. We look at three metrics: *Coverage*: the ratio of the facts that could be either supported or contradicted given the evidence; *Precision*: the ratio of supported facts to the sum of supported and contradicted facts; and finally the ratio of the *Supported* facts to all generated facts (supported, contradicted, and not enough information). By looking at Table 3, we can see that the event tiles were the most successful fact types (with  $\approx 99\%$  precision and  $\approx 72\%$  of the facts verified). This is not surprising as they are short and semantic drift can grow with length, as more noise is potentially added. The precision is so high because only 0.61% of the event tiles were contradicted. The second most successful fact type is the summary (with around  $\approx 88\%$  precision and  $\approx 58\%$  of facts supported). Again, this makes intuitive sense as these summaries focus on condensing major story beats, containing more paraphrased segments. E.g. the phrase “an evil root breeding beneath Kraghammer” from Fig. 2 is lifted almost verbatim from the transcripts. Spotlight

<sup>6</sup><https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

**Table 3: Results of the LLM Evaluator. Facts are generated from the “Title”, “Spotlight” and “Action”, and “Summary” fields of the AURA events. The 2<sup>nd</sup> to last row is results across all fact-types combined. The last row is the evaluation for events where at least 3 of the facts had the same verdict.**

| Facts              | Coverage | Precision | Supported |
|--------------------|----------|-----------|-----------|
| Event Title        | 72.39%   | 99.15%    | 71.78%    |
| Spotlight & Action | 43.56%   | 85.92%    | 37.42%    |
| Summary            | 65.64%   | 88.32%    | 57.98%    |
| Combined           | 58.28%   | 87.72%    | 51.12%    |
| Event Level        | 25.15%   | 97.56%    | 24.54%    |

and action pairs were much harder to verify. Less than half ( $\approx 44\%$ ) of these facts could be either supported or contradicted given the evidence. Because these combine two very specific type of information (character name/specific action), they can be difficult to verify if the human-made summaries focus elsewhere. The relatively high precision of these facts ( $\approx 86\%$  compared to a  $\approx 37\%$  support score) suggests that this is the case—rather than factual errors. Similarly, despite the high overall precision ( $\approx 88\%$ ), the relatively low amount of verifiable—and supported—facts ( $\approx 58.28\%$  and  $\approx 51.12\%$  respectively) can be attributed to the complexity of the “spotlight and action pair”-type facts.

In summary, the first prototype of AURA shows promising results. While many of the generated facts cannot be easily verified, only  $\approx 15\%$  of all events had 3 or more facts explicitly classified as having “not enough information”; and only 0.61% of all events had 3 or more facts classified as “contradicted”. While the coverage of events should be improved—only  $\approx 25\%$  of all events had 3 or more verifiable facts—the overall precision of the facts that can be verified is remarkably high (up to  $\approx 98\%$  for events where at least 3 facts have the same verdict).

## 7 Discussion and Limitations

In creating AURA, a number of challenges were encountered, and several key limitations remain. TTRPGs form complex narratives imposing difficulties on the automated transcription and analysis game sessions [26, 31, 42, 57, 70, 78]. While the recordings of *Critical Role* sessions are reasonable facsimiles of what a typical TTRPG session looks like, they add further complications:

- **Advertisement Breaks:** There are regular advertisement breaks in the videos, and sometimes moderators or technical staff get involved, and therefore will automatically be picked up as participants by transcription models. In a real-world TTRPG situation, such disturbances would not occur.
- **Made for Spectators:** While *Critical Role* is played as TTRPGs, the involved participants are professional actors with a monetary incentive. They are likely better at creating interesting and watchable joint narratives and supporting each other in this endeavour than a standard TTRPG group.
- **Voice Actors as Players:** The *Critical Role* cast are voice actors, and therefore better at changing their voice. This is notably the case for the GM, who, while playing different NPCs, can lead to automated transcription models picking

this up as coming from multiple people. In a practical TTRPG situation, a directional camera or separate microphones per speaker would solve this problem.

- **Ground Truth Challenges:** The ground-truth utilized here was created by fans of *Critical Role*. There might be interpretations they put on the transcripts and wiki entries. Some examples of minor data inaccuracies in the wiki entries were documented for this project.
- **Narratives for Entertainment:** The narratives in the *Critical Role* sessions are designed to entertain a broader audience, feature adult themes, and may be more complex than in TTRPG sessions run for children [6, 22, 39, 41, 46, 65].

Additional technical challenges are posed by the use of fantasy names, domain-specific knowledge (e.g., game rules), which means models created for different contexts may need to be tuned. Furthermore, all episodes are processed independently, meaning there is no consistency across episodes. This would be needed for recaps across entire campaigns. For the current instantiation of AURA, it means that player highlights are created on a per-game session (episode) basis, rather than as a function of the whole campaign (or story arc), represented in the used episodes of *Critical Role*. The solution to this challenge will be to process multi-session and single-session data in two different layers of granularity in future work.

## 8 Conclusion

The Dagstuhl Report *The Future of Games in Society* issues a call to action for research that seeks to maximise the positive societal benefit of games [28]. In response, this paper introduces AI-driven game facilitation support tools in therapeutic or other supportive contexts, and introduces a new layer of transparency and replicability. The need for this is evidenced in the global shortage of support for neurodivergent children and youth is well-documented, with widespread human and economic costs [3, 17, 21, 33, 34, 53, 60, 61, 74, 75, 77, 81, 83, 85–87, 87]. Therapeutically Assisted Role-Playing Games (TARPGs) form a promising approach towards increasing the support available for neurodivergent clients. This not only extends the toolset of therapists, but also directly allows them to work with groups of clients at a time rather than one-on-one. Furthermore, this potential societal impact can be if carers (e.g., parents, teachers) could utilize TARPGs. Even if carers cannot conduct therapy, case-based evidence suggests the impact on support for neurodivergent children could be substantial if they could utilize TARPGs [6, 10–12, 41, 45, 54, 55, 65, 66]. However, there are a number of challenges to consider, including practical barriers hindering adoption.

The AURA prototype demonstrates the feasibility of AI-driven facilitator support in group therapies, producing structured session summaries and reducing administrative burden. This has the potential to automate game monitoring and analysis tasks, which require training, resources, and take focus from the participants, and are factors in hindering the use of TARPGs not only by clinical therapists but also carers [6, 22, 39, 41, 46, 65]. Such AI-driven tools also have potential applications beyond TARPGs, informing future digital tools for group-based interventions across promotive, preventive, or therapeutic modalities.

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## A Prompts

This section presents the prompts used in the summary generation and evaluation of AURA.

### A.1 AURA Summary Generation

#### System Prompt

You are a JSON generator. You must respond ONLY with JSON that matches the provided schema. Do not include explanations.

#### Story Beat Prompt

You are given part of a tabletop RPG session transcript:

```
""{chunk}""
```

Task:

- Identify the key story beats in this chunk.
- Each beat should be 1–2 sentences, focused on story (not mechanics).
- Focus on goals, conflicts, discoveries, twists, and consequences.
- Ignore dice rolls and most jokes or small talk.
- Identify the most important character(s) in the story beat (spotlight character).
- Recap the actions of the spotlight character.

Output JSON must match this schema (already provided via the API):

```
{
  "beats": [
    {
      "number": "integer",
      "summary": "string",
      "spotlight": "string",
      "action": "string"
    }
  ]
}
```

#### Story Event Prompt

You are given story beats from several chunks of a single tabletop RPG session, in JSON:

```
{beats_json_str}
```

Task:

- Merge these beats into a single list of 3–10 major story events covering the entire session.
- Each event must have:
  - "number": integer, chronological order starting from 1

- "title": short string title
- "summary": 1–2 sentences describing what happens and why it matters.
- "spotlight": the most important character(s) during the event.
- "action": the most important action taken by the spotlight character(s).
- Focus on goals, key locations, conflicts, discoveries, twists, consequences.
- Ignore fine-grained mechanics and small talk unless they change the story.

Use the session title: "{session\_title}".

Output JSON must match this schema (already provided via the API):

```
{
  "session_title": "string",
  "events": [
    {
      "number": "integer",
      "title": "string",
      "summary": "string",
      "spotlight": "string",
      "action": "string"
    }
  ]
}
```

### A.2 AURA Evaluation

#### Evaluation Prompt

You are given a candidate event extracted from a story recap and a human-written source describing the same tabletop RPG session.

EVENT (to verify):  
""{fact}""

SOURCE EVIDENCE (human-written recap):  
""{relevant\_paragraphs}""

Task:

Verify whether the event is supported by the evidence.

Follow these rules:

- Use information stated in the evidence.
- No assumptions, inference, or world knowledge.
- No strengthening or embellishing claims.
- Focus on the action and events.
- Disregard surface-level descriptions and adjectives.

Provide justification for each decision and cite exact spans from the evidence.

- Use literal quotes, not paraphrases.
- Do not include any information not present in the evidence.
- Do not infer motivation, chronology or relationships unless explicitly stated.
- ALWAYS provide the evidence from the source if there is no evidence, it is a contradiction.

Choose exactly one label:

- SUPPORTED: The event is supported by the evidence.
- CONTRADICTED: The event is not supported by the evidence.
- NOT\_ENOUGH\_INFORMATION: There is not enough information to prove or disprove the event.

Output JSON must match this schema (already provided via the API):

```
{
  "verdict":
    "SUPPORTED |
    CONTRADICTED |
    NOT_ENOUGH_INFORMATION",
  "justification": "string",
  "evidence": "string"
}
```

## B Example Output

### B.1 Story Beats

Example output for the first chunk (3000 words) of the first session.

**Chunk: 1**

**Beats:**

**Number: 1**

**Summary:** A dark vision has come to Lady Kima, a halfling paladin of Bahamut, warning of an evil root breeding beneath Kraghammer and the surrounding mountains.

**Spotlight:** Lady Kima

**Action:** She shares this information with Allura Vysoren of the Tal'Dorei Council, who requests their aid in finding Lady Kima and investigating the source of the dark vision.

**Number: 2**

**Summary:** The party is ambushed by a group of roaming barbarian goliaths on the pathway to Kraghammer, but Grog uses his social encounter skills to avoid battle.

**Spotlight:** Grog

**Action:** He rolls well on a persuasion check and manages to talk

his way out of combat, allowing the party to continue their journey.

**Number: 3**

**Summary:** The party gains entry into Kraghammer after presenting their paperwork and is greeted by the city's front guards.

**Spotlight:** None

**Action:** No specific action taken by a spotlight character, but the party begins to explore the city.

**Number: 4**

**Summary:** The party discovers that Kraghammer is an enormous underground metropolis with a unique architecture and a dark, ominous atmosphere.

**Spotlight:** None

**Action:** No specific action taken by a spotlight character, but the party begins to explore the city.

### B.2 AURA Events

Example of the events generated for the first session.

#### A Dark Vision and a Plea for Help

**Summary:** Lady Kima, a respected halfling paladin of Bahamut, warns the party of an evil root breeding beneath Kraghammer and asks for their aid in finding her whereabouts.

**Spotlight:** Lady Kima

**Action:** She requests the aid of the party in finding her whereabouts and bringing her back safely.

#### Ambush at the Roadside

**Summary:** The party is ambushed by a group of roaming barbarian goliaths, but Grog recognizes one of them as an old ally and manages to avert battle through his social encounter.

**Spotlight:** Grog

**Action:** He recognizes one of the goliaths as a previous ally and manages to avert battle.

#### Arrival in Kraghammer

**Summary:** The party arrives at the city of Kraghammer, meets Adria, and learns about Lord Greyspine, the city's Iron Keeper.

**Spotlight:** Adria

**Action:** She provides information about Lord Greyspine and warns them about potential dangers in the lower levels.

#### Investigating House Thunderbrands

**Summary:** The party asks Adria about mystics studying arcane arts, and she points them to House Thunderbrands.

**Spotlight:** Adria

**Action:** She also mentions that there have been troubles at the Greyspine quarry due to goblin activity.

#### Meeting with Lord Greyspine

**Summary:** The party learns more about Lord Greyspine and his role in the city, including a potential meeting with him.

**Spotlight:** Nostoc (party member)

**Action:** They ask Adria to help them find a way to meet with Lord

Greyspine.

#### Grog's Acquisitions

**Summary:** Grog and the party acquire valuable items, including thistle blood wine, from a barmaid in exchange for their services.

**Spotlight:** Grog

**Action:** He negotiates with the barmaid to trade a troll deck for the wine.

#### Exploring Kraghammer's Residential Districts

**Summary:** The party explores the city, looking for information about the Greyspine Quarry and discovers that it is expanding into the lower levels.

**Spotlight:** Pike

**Action:** He uses Dimension Door to blink through a small doorway and emerges outside with Scanlan.

#### The Sealed Entrance

**Summary:** The party discovers a sealed entrance to Kraghammer, guarded by dwarves who are waiting for something to come through.

**Spotlight:** Party (general)

**Action:** Keyleth uses her inspiration to inspire Vax, Grog, and Tiberius as they prepare for battle.

#### The Dark Tunnel

**Summary:** A dark, venomous creature emerges from a tunnel, but the party defeats it with Scanlan's lightning bolt.

**Spotlight:** Scanlan, Keyleth, Tavirius

**Action:** Scanlan's lightning bolt kills the creature.

**Evidence:** The source evidence states: 'Vox Machina heads back to the city to rest and stock up, but first they visit House Thunderbrand.'

**Justification:** The event mentions visiting the House of Arcane Practicing and Study, but the source evidence mentions visiting House Thunderbrand.

**Verdict:** Contradicted.

**Fact:** Spotlight character: Grog. Main action: He negotiates with the barmaid to trade a troll deck for the wine.

**Evidence:** No specific evidence found.

**Justification:** The event mentions Grog negotiating with a barmaid to trade a troll deck for wine. However, there is no mention of this event in the provided source evidence. The source evidence does mention Grog and interactions with a tavern, but it does not specifically mention a barmaid, a troll deck, or a trade for wine.

**Verdict:** Not enough information.

## B.3 Evaluation Output

Examples of the "Supported", "contradicted", and "not enough information" verdicts.

**Fact:** Spotlight character: Lady Kima. Main action: She requests the aid of the party in finding her whereabouts and bringing her back safely.

**Evidence:** Vox Machina have just entered the dwarven city of Kraghammer for the first time on their quest to find Lady Kima of Vord.

**Justification:** The event states that Lady Kima is the spotlight character and the main action is that she requests aid in finding her whereabouts and bringing her back safely. The source evidence mentions that Vox Machina are looking for Lady Kima, as they ask Adra about a halfling and learn that Lady Kima was headed for the mithral mine owned by Nostoc Greyspine. This indicates that the party is indeed seeking her and trying to find her location, which aligns with the event.

**Verdict:** Supported.

**Fact:** The party visits the dwarven House of Arcane Practitioning and Study, seeking guidance on a dark arcane entity in one of their mines, but Tiberius Stormwind declines to help.